

. DEVELOPED BY TEAM PULSE .

Lunar PSR Image Enhancer

Enhancement of Permanently Shadowed Regions (PSR) of Lunar Craters Captured
by OHRC of Chandrayaan-2

The Problem of Lunar Shadows

- **Permanently Shadowed Regions (PSRs)** near the lunar poles receive zero direct sunlight.
- **Extreme contrast** makes raw images pitch black, completely hiding dangerous rocks and topography.
- **Critical hazard** for unlit lunar rovers and safe landing site selection.

OUR SOLUTION

Lunar PSR Image Enhancer

Simple Interface

A clean, interactive dashboard that anyone can use easily.

Target Users

Designed specifically for space researchers and autonomous mission planners.

Performance

Delivers instant image enhancement and hazard mapping in under 3 seconds.

HOW IT WORKS

Our Three-Step Process

1

Noise Removal

Cleans up heavy camera static without blurring the rocks.

2

Smart Brightening

Uses math to reveal details hidden inside pitch-black shadows.

3

Hazard Mapping

Automatically finds craters and highlights dangerous landing spots.

LIVE DEMONSTRATION

Image Enhancement & Metrics

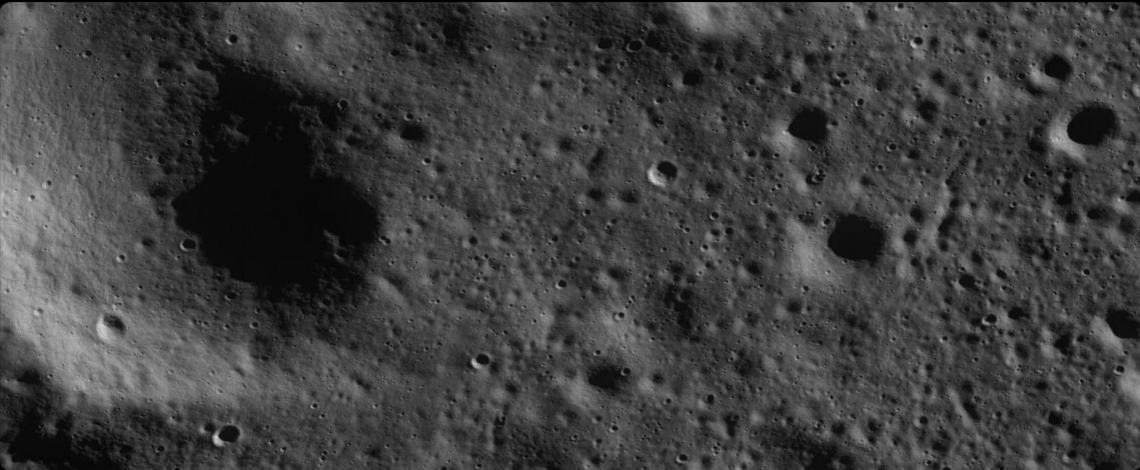
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Image Enhancement Crater & Hazard Detection Upload New Image

Original Dark Image



Enhanced Image



Image Brightness

121.29

↑ 57.15

Detail Score (Entropy)

7.64

↑ 0.78

Hazards / Craters Found

35

↑ Tracking Active

Download Enhanced Image

Automated Crater Detection Dashboard

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Automated Crater Detection

Hover over the highlighted areas to view the size and circularity of the detected craters.

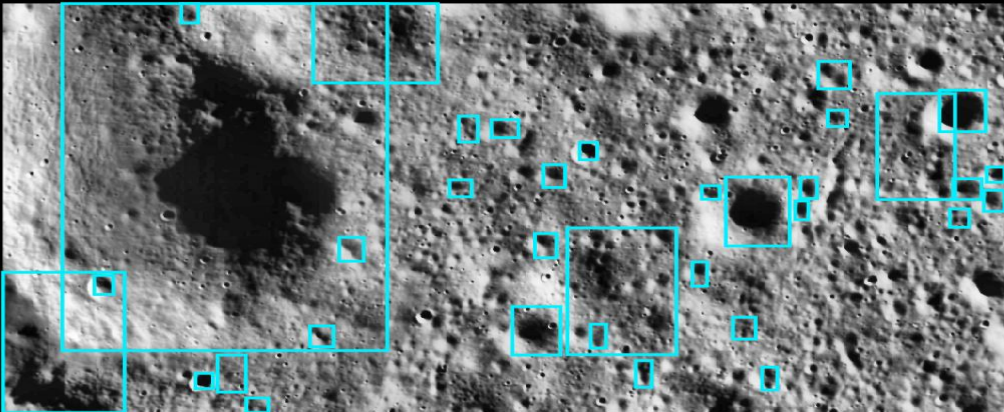


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WHY IT MATTERS

Helping Future Space Missions

- **Unlocks hidden data** successfully from Chandrayaan-2 OHRC imagery.
- **Scalable framework** fully designed for future lunar south pole missions and autonomous rovers.

Thank You

We are ready for your questions.

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TEAM PULSE
SAMARTHYA AND ARYAN
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